Lappeenranta-Lahti University of Technology LUT
School of Engineering Science
Computational Engineering and Technical Physics
Computer Vision and Pattern Recognition

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IMAGE LEVEL MICRO GESTURE CLASSIFICATION

Practical Assignment

Examiners: Rong Gao
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ABSTRACT

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The objective of this project involved working with the 32 micro-gesture (MG) categories using images extracted from the iMiGUE dataset established by the Center for Machine Vision and Signal Analysis at the University of Oulu. The aim of this project is to classify 32 micro-gesture (MG) categories using images extracted from the iMiGUE dataset provided by the Center for Machine Vision and Signal Analysis of the University of Oulu, Finland. In this project, we have designed a system that detects the gestures/poses present from the video frame, preserving actors' facial identities, and classifies the gestures based on the 32 MGs. We have used a Bidirectional Long Short-Term Memory (BiLSTM) network along with MediaPipe Holistic to build the model and train over the processed dataset. The pipeline abstracts raw images into skeleton keypoints; the points' temporal dynamics are processed by BiLSTM. During the project implementation, some challenges were encountered in handling the data imbalances along with choosing the right algorithms to proceed with the classification.

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1 Project Collaboration

This project has been executed solo. During the inception of the project, I designed the project roadmap and solely executed all the tasks—from brainstorming and ideation to implementation and report writing.

- Exploring the problem statement and reading related papers and articles on Mediapipe
- Initial ideation and project complexity analysis to ensure I am not overloaded while equally ensuring the quality of the model is intact.
- Implemented image preprocessing codebase.
- Evaluate model performance based on the trained model.
- Evaluation, visualization, and report writing.

2 Introduction

2.1 Background

Micro gestures classification and identification have been challenging tasks in computer vision. Although it carries significant impacts and has wider implications, the underlying problems to achieve its full potential has not been achieved. Gesture classification helps to read actors' emotional or psychological nonverbal signals, express their opinions through universal gestures, and plays an important role in human-computer interactions. Historically, the classification heavily relied on analyzing RGB video streams using complex 3D CNNs, making it computationally expensive and difficult to get quality large datasets. To mitigate these challenges, skeleton-based pose and gesture detection has been widely adopted as it captures essential kinematic and geometric features. The iMiGUE dataset prepared by the Center of Machine Vision and Signal Analysis at the University of Oulu provides a collection of 32 labeled micro-gesture classes, images/gestures recorded at a sporting event, for pattern recognition research and better understanding of emotion from the gestures and pose. In this project we explore the dataset to build and test a model that accurately ingest the RBG video streamns skeletion points in 3D space and predicts the gestures based on those skeletion points in the temporal space.

2.2 Objectives and delimitations

This project implements an image-level microgesture classification pipeline by integrating Google's MediaPipe Holistic to extract 225 accurate 3D landmarks that capture hands and body poses. It utilizes RGB images from the iMiGUE dataset, processes and extracts skeletal data, builds a robust neural network pipeline for training, and rigorously tests the model's performance.

The primary objectives of this project are broken down into the following key areas:

- **Efficient Feature Extraction:** To use MediaPipe Holistic to sort raw, pixel-heavy image frames into lightweight, 225-dimensional 3D skeletal landmark vectors that capture both hands and face.
- **Temporal Modeling:** To design and train a Bidirectional Long Short-Term Memory (BiLSTM) neural network capable of learning the sequential and temporal dy-

namics of the extracted skeletal movements.

- **Class Imbalance Resolution:** Use of focal loss during training of the model to ensure model performance across all 34 gestures is not hampered.

To maintain a focused and computationally feasible scope, the model utilizes a fixed temporal window of exactly 16 frames per sequence. Furthermore, the pipeline strictly evaluates the BiLSTM architecture without benchmarking against other temporal models. The feature space bounds only to the 225 body and hand coordinates and ignores facial landmarks or environmental background variables.

2.3 Project Structure

This section gives a glimpse of project artifacts and the structure of implementation. The below-mentioned files show how the implementation has been carried out and what each folder holds for the project.

`config.py` and `utils.py`: `config.py` Both of the scripts manage the variables and parameters shared across the model pipelines. centralizes all global hyperparameters, such as the fixed sequence length (16 frames), the expected feature dimensions (225), class limits, and training variables (learning rate, batch size, epochs count).

`utils.py` provides shared helper functions for chronological frame sorting, sequence normalization, setting dataset paths, and all other supporting functions.

`preprocess_dataset.py`: This script deals with preprocessing of the original dataset and splits the processed data into 80/20 train-test subsets. It also enforces the strict temporal window by uniformly downsampling or padding every video sequence to exactly 16 frames.

`extract_skeletons.py`: This file iterates through the preprocessed frames and passes them through the Google MediaPipe Holistic framework. It extracts the 3D coordinates of the pose and both hands, concatenating them into a flat, 225-dimensional .npy vector.

`train_model.py` This artifact contains the PyTorch implementation of the Bidirectional Long Short-Term Memory (BiLSTM) network. It utilizes an AdamW opti-

mizer and focal loss to penalize misclassifications and further improves the model accuracy on each epoch.

`test_trained_model.py` This module handles the rigorous testing of the finalized model against the unseen 20% test split. It calculates and plots the confusion matrix and F1 scores for each class during the evaluation phase.

3 Related Work

Micro-gesture recognition is a complex task because the movements are often very short and subtle. This makes it harder for typical models to detect and classify them accurately. The iMiGUE dataset introduced by Liu et al. [1] and hosted openly [2] includes 32 micro-gesture classes recorded in real-life situations. From their works and our initial understanding, handling these type of data requires a model that deals with vision. For classification of gestures and to capture all, we need to build an architecture that handles appearance and motion features. Earlier versions use 3D Convolutional Neural Networks (CNNs) that were resource intensive and required heavier qualitative datasets. With an advent of Google MediaPipe, we can handle all the classes (32) by converting video labels into skeletons.

ResNet, introduced by He et al. [3], includes skip connections that help it train deeper networks more effectively and has been used in this project to ensure learning from visual input.The motion patterns contained within pose data can be studied by our modified neural network through its LSTM layer. Inside Zhang et al. [4] an RGB information and pose information method has been discussed that we have further used to extract skeletons from the labels. We implemented a hybrid approach to build architecture for the project.

4 Proposed Methods

In this section we describe the rationale behind the chosen methods. We discuss algorithmic choice, data processing, and training procedures.

4.1 Algorithms

The project follows a multi-stage algorithmic pipeline designed to transform raw video data into skeleton points that are used for gesture classification.

1. **Temporal Sampling Algorithm:** Standardizes variable-length video sequences into a fixed temporal window of exactly 16 frames to satisfy the static input requirements of the recurrent neural network.
2. **Skeletal Extraction Algorithm:** Utilizes the MediaPipe Holistic framework to process raw RGB pixels and output sparse, 3D coordinate graphs of the subject's pose and hands. We focused on using "holistic" instead of "pose" as it covers both hands and gestures instead of pose only.
3. **Spatial Normalization Algorithm:** This anchors the extracted floating skeleton's coordinates to relative reference joints, making the data invariant to the subject's absolute position within the frame.
4. **Sequence Classification Algorithm:** A Bidirectional Long Short-Term Memory (BiLSTM) network processes the temporal sequence of normalized graphs to predict the final MG class.

4.2 Data Processing

The data processing for the project has been handled in two major phase; processing data into .npy and preparing skeleton data for the model training. The iMiGUE dataset is first capped, restricting any single gesture class to a maximum of 500 video sequences to reduce the class imbalance. The data are processed into train and test upfront before conversion into skeleton keypoints. Similarly, classes having less labels are integrated into training set.

Once the frames are sampled, MediaPipe Holistic extracts 3D landmarks (x, y, z). The pipeline isolates 33 pose landmarks, 21 left-hand landmarks, and 21 right-hand landmarks. These are concatenated into a flat 225-dimensional feature vector per frame. Sequences where the algorithm fails to detect a skeleton in more than 80% of the frames are discarded as dead data.

The normalization is applied based on anchor joints:

- **Pose:** Normalized relative to the nose (index 0).
- **Left Hand:** Normalized relative to the left wrist (index 33).
- **Right Hand:** Normalized relative to the right wrist (index 54).

4.3 Model Architecture

The MG classification is performed by a bidirectional LSTM (BiLSTM). Let the input sequence be defined as $X = (x_1, x_2, ..., x_T)$, where the sequence length is 16 frames and $x_t \in R^{225}$.

The temporal encoder consists of a 2-layer BiLSTM with a hidden state size of 128. By processing the sequence in both forward and backward directions, the network captures contextual motion dynamics from the past and the future. A dropout rate of 0.3 is applied between the LSTM layers to mitigate overfitting.

4.4 Training

The network is trained over 30 epochs with a batch size of 32. Optimization is handled by the AdamW optimizer, utilizing a learning rate of 10^{-3} and a weight decay coefficient of 10^{-4} to penalize large model weights.

The network is optimized using focal loss in place of cross entropy (CE) to handle class imbalances during the training. E. Focal loss dynamically scales the CE loss based on prediction confidence and penalizes the misclassification examples.

5 Experiments

This section details the experimental setup, implementation specifics, and the evaluation metrics used to assess the performance of the proposed skeleton-based BiLSTM pipeline for micro-gesture classification.

5.1 Implementation Details

The pipeline is implemented in Python. Skeletal feature extraction is performed using the Google MediaPipe Holistic and OpenCV libraries. The BiLSTM model and training routines are built using the PyTorch deep learning framework.

All model training and inference scripts are designed to automatically utilize CUDA-enabled GPUs when available to accelerate tensor computations. The network is trained for 30 epochs using a batch size of 32. The training loop evaluates the model against the validation split at the end of each epoch, strictly saving the model weights only when a new maximum validation accuracy is reached.

5.2 Evaluation Metrics

To comprehensively evaluate the classification performance across all 34 classes, the best-performing model checkpoint is evaluated on the 20% holdout test set. The primary evaluation metrics include:

- **Accuracy:** The overall percentage of correctly classified micro-gesture sequences across the entire test set.
- **Precision, Recall, and F1-Score:** Calculated per class using the `scikit-learn` library to provide granular insights into the model's performance, particularly highlighting how it handles classes with overlapping or subtle kinematic similarities.

6 Results and Discussion

This section presents the quantitative and qualitative results of the BiLSTM micro-gesture classification model evaluated on the 20% holdout test set.

6.1 Overall Classification Performance

Our model achieved an overall test accuracy of **60.87%** across all 32 micro-gesture classes. Using Google MediaPipe's solutions framework for the landmarks and combining it with a BiLSTM to train the model, the architecture was able to achieve a moderate overall performance, which is reflected by a mean F1-score of 0.57.

6.2 Class-Specific Analysis

To understand the model's granular performance, precision, recall, and F1 scores were computed for each distinct micro-gesture. Figure 1 illustrates the F1-score distribution across all 32 classes. The model demonstrated high confidence and accuracy for several gestures, peaking with Class 3 (F1: 0.93) and Class 14 (F1: 0.91).

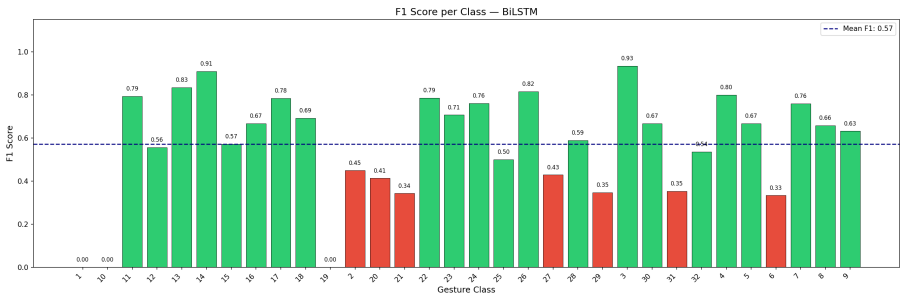

Figure 1. Per-class F1-score distribution for the 32 micro-gesture classes, indicating a mean score of 0.57.

Conversely, some classes, such as Class 1, 10, 19 with F1-score of 0.00. Because the spatial normalization anchors all coordinates to the wrists, micro-variations in individual finger joints occasionally lacked the temporal variance required for the BiLSTM for classification.

6.3 Confusion Matrix and Misclassification Analysis

To further diagnose the model's predictive behavior, a confusion matrix was generated (Figure 2). The pronounced diagonal confirms that a large portion of predictions align precisely with the ground truth labels, particularly for the highest-performing classes.

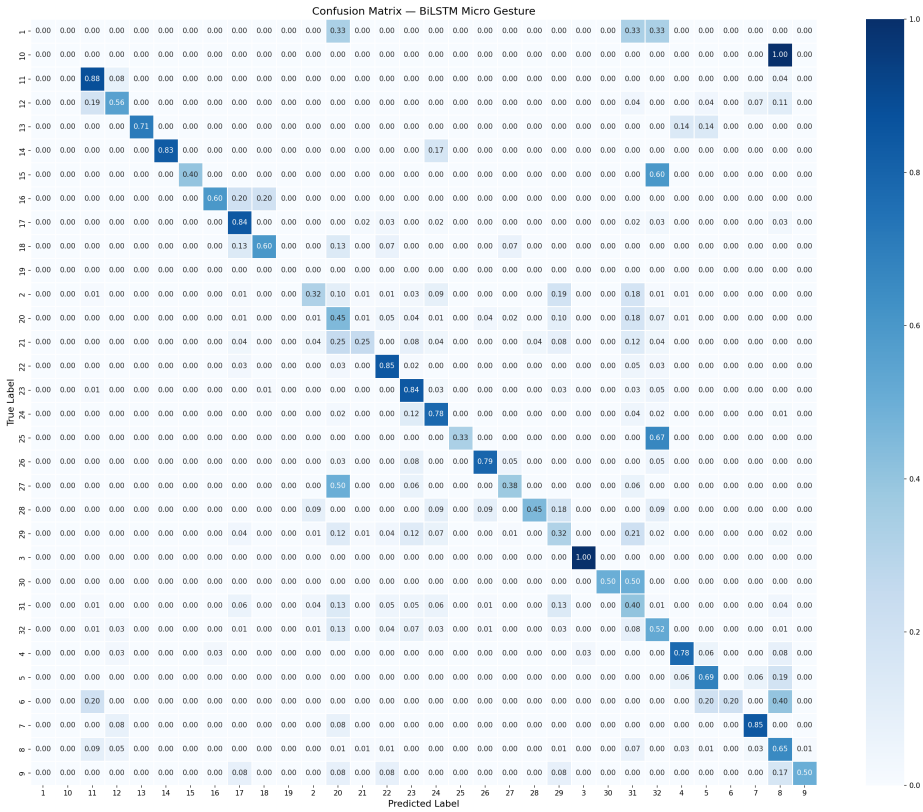

Figure 2. Confusion Matrix of the BiLSTM model predictions on the test set.

Analysis of the off-diagonal elements reveals specific kinematic overlaps between certain gesture pairs. For instance, there is a total misclassification rate of **100%** where the model incorrectly predicts Class 10 as Class 8. Similarly, predictions for Class 30 are split evenly, misclassifying it as Class 31 (**50%**) and Class 32 (**50%**). This confusion is geometrically logical, as these gesture pairs likely share a nearly identical starting pose and hand shape, differing only in the final frames of the sequence.

7 Conclusion

This thesis presented a robust, end-to-end machine learning pipeline for classifying 34 distinct micro-gestures from the iMiGUE dataset. By choosing an architecture that offers computationally less intensive and robust, Google's MediaPipe holistic framework and BiLSTM are combined.

The integration of spatial normalization—anchoring dynamic joint coordinates to relative root points such as the nose and wrists—ensured that the model remained unchanged by background clutter, camera angles, and subject positioning. Furthermore, the application of a Bidirectional Long Short-Term Memory (BiLSTM) network proved highly effective at capturing the continuous temporal dynamics of these sequences, processing the geometric data in both forward and backward directions to understand the full context of the microgestures.

Additionally, to deal with challenges of real-world datasets and imbalances To address the inherent challenges of real-world dataset imbalances and the subtle nature of micro-gestures Focal loss during training is implemented. Ultimately, this project demonstrates that sparse skeletal representations, when paired with recurrent temporal modeling, offer a computationally efficient and highly accurate approach to identity-free gesture recognition.

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